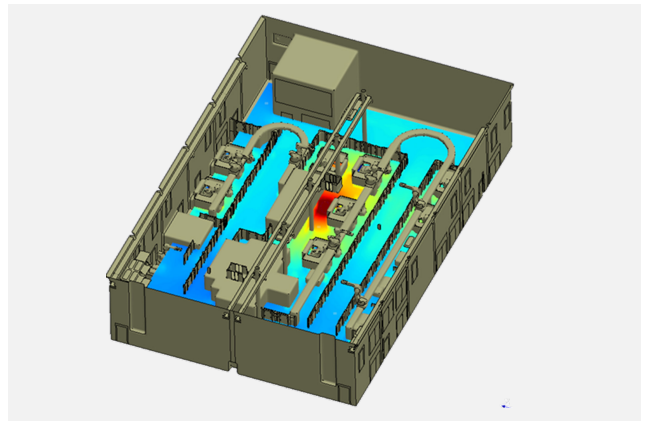
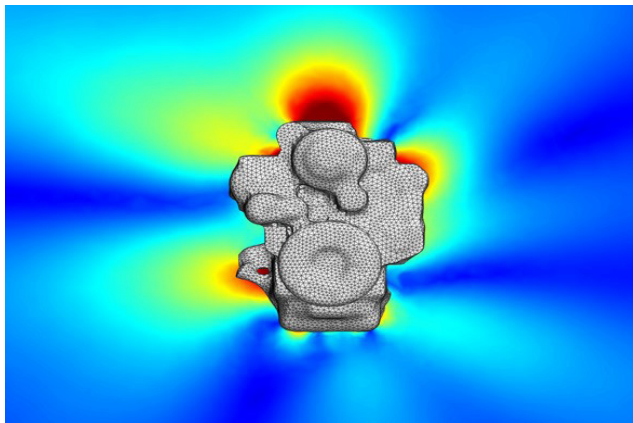


Actran™

Powerful acoustic simulation software

The Actran software suite is intensively used by leading companies in automotive, aerospace, consumer electronics, and other industries to improve the acoustic performance of their products. Actran offers general-purpose acoustic, vibroacoustic, and aeroacoustic capabilities. Thanks to its extensive modelling capabilities, intuitive graphical user interface, and powerful solver implementation, Actran is the ideal companion for engineers who need to simulate advanced acoustic problems easily, efficiently, and accurately.



Actran Acoustics

Your solution for interior and exterior acoustic propagation, weakly coupled vibroacoustics, and duct acoustics

Actran Acoustics is the core module of Actran. It contains a wide range of modelling features, allowing users to predict interior and exterior noise propagation problems. Actran Acoustics can account for complex effects such as flow and temperature effects, absorbing materials, visco-thermal losses, and more. Actran includes time domain and frequency domain solvers, and interfaces with structural FEA (such as MSC Nastran™) and multibody dynamics solvers (Adams™) to compute the acoustic radiation from vibrating structures.

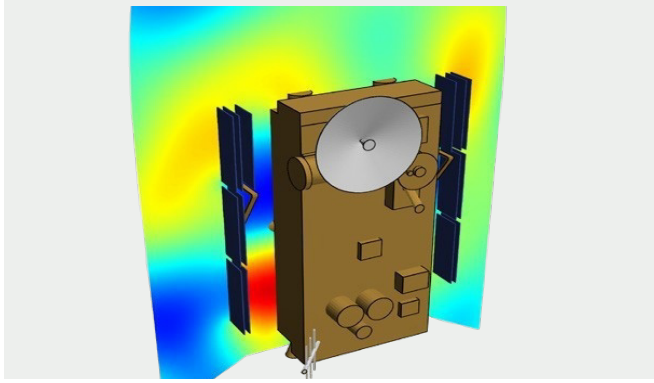
Examples of application: acoustic radiation of vibrating structures such as internal combustion or electric motors, gearbox and compressors; duct acoustics (mufflers, silencers, air conditioning units, distribution systems); pass-by noise

Actran Radiosity

Your solution for interior and exterior space acoustic propagation at high frequency

Actran Radiosity is a module dedicated to open and/or closed heterogeneous spaces, enabling high-frequency noise propagation based on the acoustic energy radiative transfer assumption in the frequency domain. It suits well to large-space applications, but is not restricted to them. Actran Radiosity can account for diverse noise source definitions from classical monopole sources to panel vibrations derived from simulation and/or measurement. Thanks to its solver architecture, Actran Radiosity provides results in an extremely fast way – from a couple of hours to less than a minute. Diverse boundary conditions can be applied to walls, from purely reflective to purely absorbent.

Examples of application: large-space room acoustics from factories, rooms, houses, warehouses, etc.; cabin noise at high frequencies from vehicles, aircraft, railways; environmental noise propagation

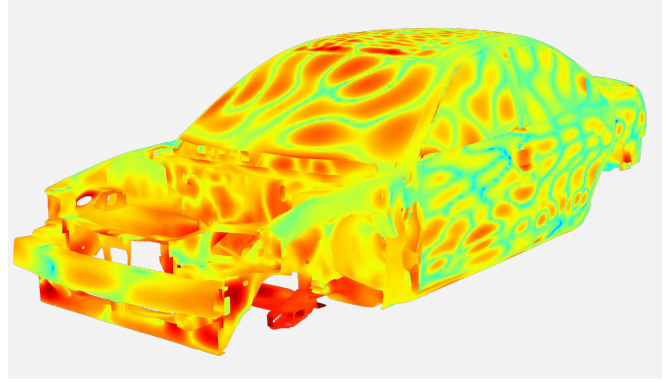


Actran Vibroacoustics

Your solution for strongly coupled vibroacoustics

Actran Vibroacoustics extends the features of Actran Acoustics with capabilities to solve structure dynamics and the interactions between a structure and an acoustic fluid. Structures are modelled directly in Actran or represented by their modes computed by an external FEA solver. In addition, users can introduce complex absorbing materials, such as foams, into the model. Structures can be loaded with excitations representing the most realistic conditions, from simple loadings to diffuse sound field or turbulent boundary layer stochastic excitations.

Examples of application: sound transmission through panels (windows, sealing, cockpits, fuselages); acoustic fatigue of spacecraft structures; vibroacoustic strong interactions (loudspeakers, submerged structures, piezo-electric actuators)

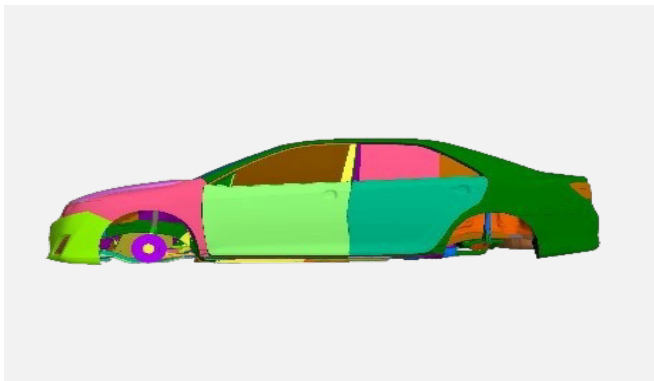


Actran for Trimmed Body

Your solution for NVH of trimmed vehicle bodies

Actran for Trimmed Body is a powerful tool to model and analyse the effect of acoustic trim components on complex structures. Structure models computed by FEA solvers such as MSC NastranTM are complemented by a detailed model of acoustic trim components to account for structural damping, insulation, and acoustic absorption. Actran for Trimmed Body allows users to accurately predict the noise perceived by passengers from any kind of vibration (e.g., engine or suspension mounts) or noise source (e.g., wind noise, exhaust, or engine airborne propagation).

Examples of application: NVH analysis of car trimmed body; in-flight passenger comfort; truck cabin noise

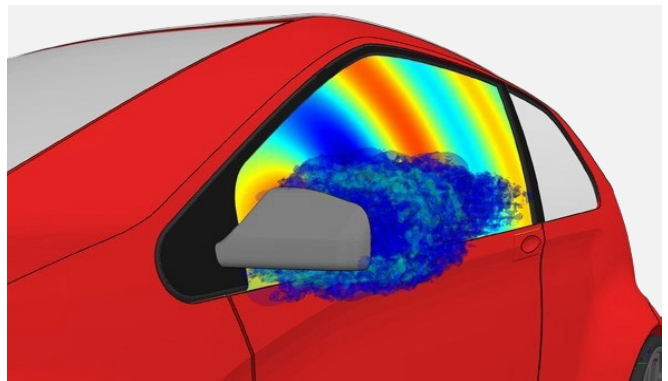


Actran SEA

Your solution to extend your vibro-acoustic analysis range

The Actran SEA module offers the possibility to extend the usability of existing vibroacoustic finite element models to mid and high frequencies. Based on automatic or user-defined subsystem definitions, the SEA parameters are efficiently extracted from the finite element model. Even without SEA expertise, sound and vibration analysis at mid and high frequencies can be performed along with energy transfer path analysis.

Example of application: boat noise onboard prediction; vehicle transfer path analysis; interior acoustic comfort

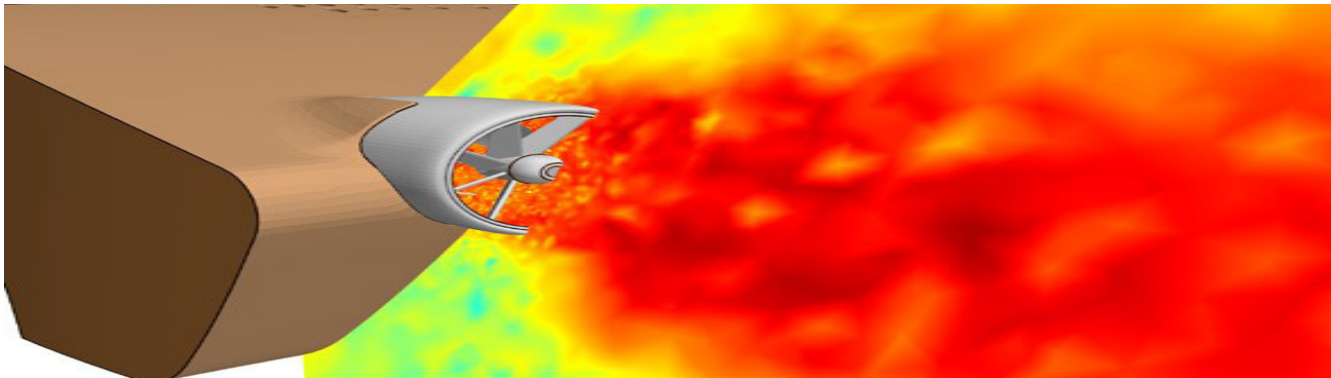


Actran Aeroacoustics

Your solution for predicting flow-induced noise

Actran Aeroacoustics features advanced capabilities to accurately predict noise generated by turbulent flows. Based on unsteady CFD flow simulation results, Actran Aeroacoustics computes the equivalent noise sources and their acoustic propagation. Actran can interface with scFLOWTM, ANSYS FluentTM and CFXTM, StarCCTM, PowerflowTM, OpenFOAMTM and others.

Example of application: flow noise in air conditioning or exhaust systems; side mirror noise; noise generated by rotating axial or centrifugal fans; airframe noise (landing gear, trailing edge)

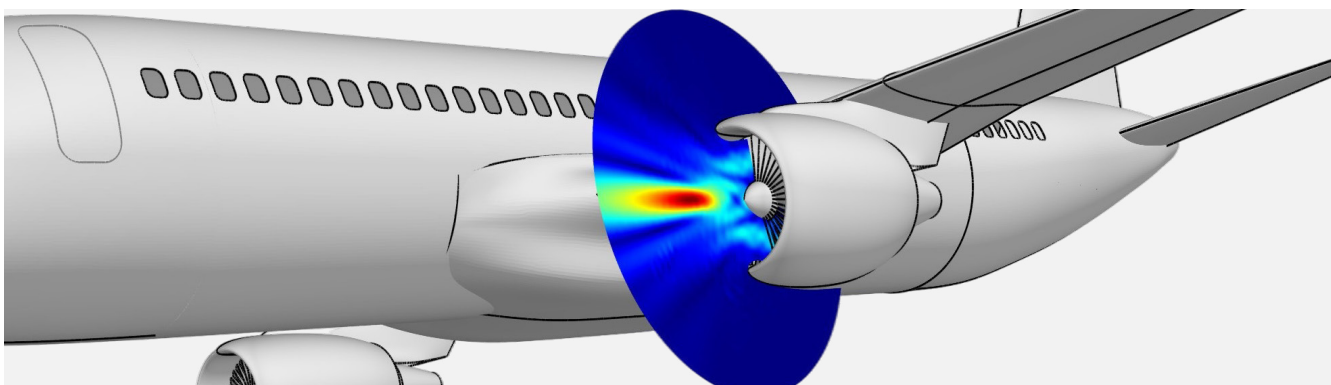


Actran SNGR

Your solution for fast flow noise computation

Actran SNGR features innovative capabilities to predict the noise generated by turbulent flows in a much faster way than with classical aeroacoustic approaches. It uses RANS CFD analysis results to synthesise the aeroacoustic noise sources, which allows for a drastic reduction of the complete calculation process. Actran SNGR is particularly useful when relative levels between different designs are needed, such as in an optimisation process.

Example of application: flow noise in air conditioning or exhaust systems; side mirror noise; airframe noise (landing gear, trailing edge); wind noise around the complete car body



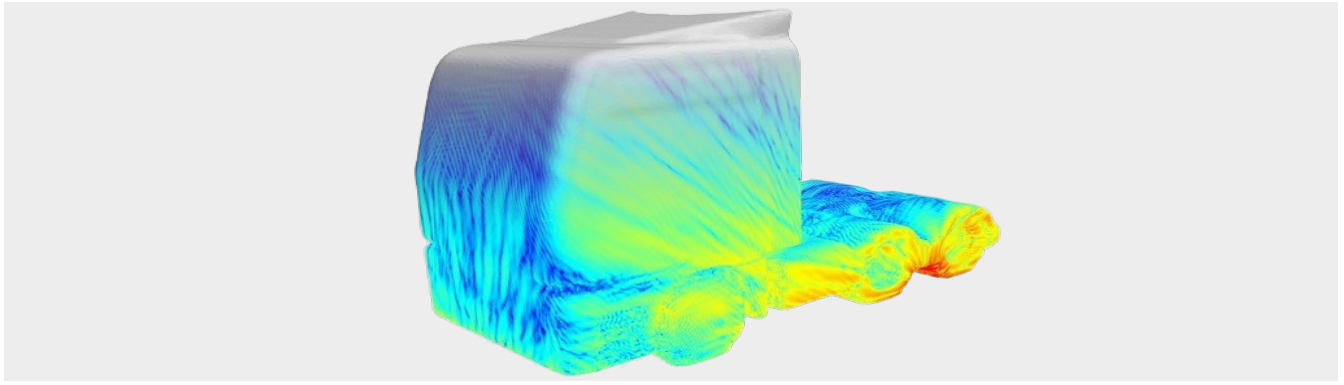
Actran TM

Your solution for turbomachinery noise

Actran TM is the aerospace industry standard to analyse the sound radiated by turbomachinery and optimise the related acoustic treatments. One of the challenges of acoustic CAE methods is handling large models associated with high wave numbers and large geometrical sizes and complexities. Actran TM provides efficient solver technologies and advanced parallel processing to address this challenge. Actran TM can be complemented by Actran DGM to solve problems involving complex shear layers and flow gradients occurring at the engine exhaust.

Example of application: optimisation of acoustic treatments for aircraft turbofan engine inlets or helicopter turbines; high-frequency propagation in duct systems (ducted cooling systems, car exhaust)



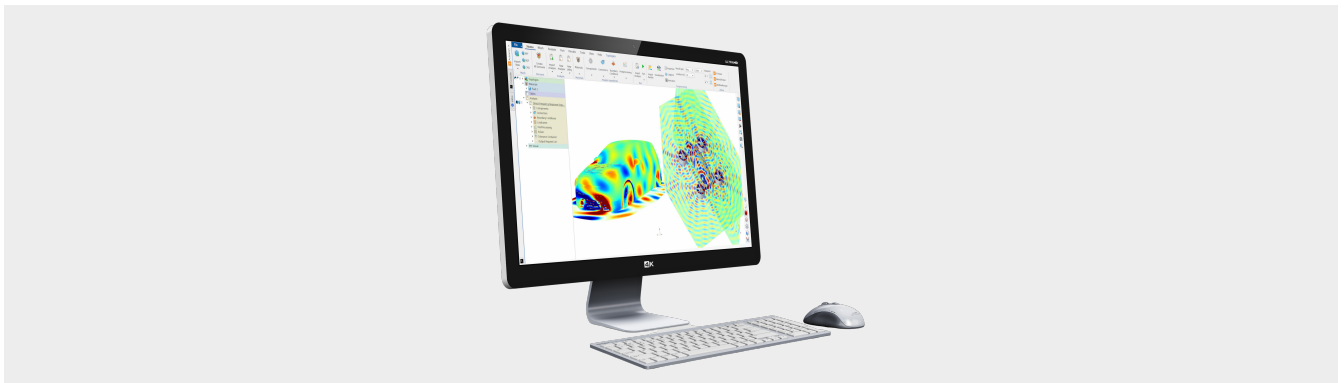


Actran DGM

Your solution for acoustic radiation in large domains, complex flows, and high frequencies

Actran DGM solves the linearised Euler equations in the time domain with the discontinuous Galerkin method. This grants Actran DGM the unique capability to solve acoustic propagation in complex flows such as high temperature gradients, shear layers, or supersonic flows. With its adaptive high-order elements, remarkable scalability, and GPU acceleration capabilities, Actran DGM is particularly well suited for studying acoustic radiation from large structures or at high frequencies.

Example of application: noise propagation at the exhaust of turbomachines; ultrasonic sensors; noise radiation from large structures (complete vehicle bodies, wind turbines)



Actran VI

Dedicated pre- and post-processing for the Actran suite

The Actran software suite is supported by an intuitive graphical user interface (GUI) dedicated to Actran model pre- and post-processing. Actran VI features its own meshing tools specifically designed for generating, modifying, and improving meshes for vibro- and aero-acoustic analyses starting from any existing CAD or mesh file. Actran VI also includes pre-processing tools to ease the creation and editing of any Actran model. A large variety of post-processing tools, such as plots, maps, animations, and waterfall diagrams, is also available to help users extract the most information out of each calculation. The Actran GUI and solver are thoroughly supported by a Python API, which makes automation or optimisation accessible to any user.

Take the acoustic performance of your products to the next level with a complete simulation solution used by leading companies around the world!